In [1]:
```
import pandas as pd
import numpy as np
```

In [3]:
```
data={
    'Age':[22,25,35,40,28,45,23,38,30,50],
    'Income':[25,30,70,80,45,90,20,75,50,100],
    'Student':[1,1,0,0,1,0,1,0,1,0],
    'Buy':[0,0,1,1,0,1,0,1,1,1]
}
```

In [5]:
```
data
```

Out[5]:
```
{'Age': [22, 25, 35, 40, 28, 45, 23, 38, 30, 50],
 'Income': [25, 30, 70, 80, 45, 90, 20, 75, 50, 100],
 'Student': [1, 1, 0, 0, 1, 0, 1, 0, 1, 0],
 'Buy': [0, 0, 1, 1, 0, 1, 0, 1, 1, 1]}
```

In [9]:
```
df=pd.DataFrame(data)
```

In [11]:
```
Y=df[['Buy']]
X=df.drop(labels=['Buy'],axis=1)
```

In [13]:
```
Y
```

Out[13]:

	Buy
0	0
1	0
2	1
3	1
4	0
5	1
6	0
7	1
8	1
9	1

In [15]:
```
X
```

Out[15]:

	Age	Income	Student
0	22	25	1
1	25	30	1
2	35	70	0
3	40	80	0
4	28	45	1
5	45	90	0
6	23	20	1
7	38	75	0
8	30	50	1
9	50	100	0

Split Data

In [18]:
```
from sklearn.model_selection import train_test_split
xtrain,xtest,ytrain,ytest=train_test_split(X,Y,test_size=0.2,random_state=42)
```

Create Model

In [49]:
```
from sklearn.tree import DecisionTreeClassifier
dt=DecisionTreeClassifier(criterion='entropy',)
```

In [51]:
```
model=dt.fit(xtrain,ytrain)
```

Prediction

In [54]:
```
y_pred=model.predict(xtest)
```

In [56]:
```
y_pred
```

Out[56]:
```
array([0, 0], dtype=int64)
```

Evalution

In [59]:
```
from sklearn.metrics import accuracy_score,confusion_matrix,classification_report
accuracy_score(ytest,y_pred)
```

Out[59]:
```
0.5
```

In [61]:
```
confusion_matrix(ytest,y_pred)
```

Out[61]:
```
array([[1, 0],
       [1, 0]], dtype=int64)
```

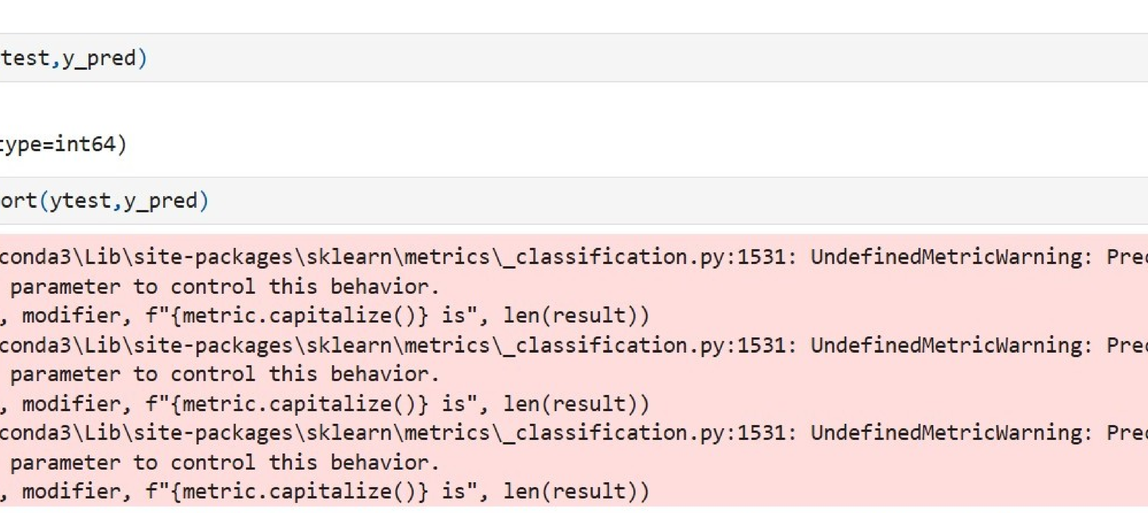
In [63]:
```
classification_report(ytest,y_pred)
```

```
C:\Users\Lenovo\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1531: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples.
Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
C:\Users\Lenovo\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1531: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples.
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C:\Users\Lenovo\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:1531: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in labels with no predicted samples.
Use `zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

Out[63]:
```
'              precision    recall  f1-score   support\n\n           0       0.50      1.00      0.67         1\n           1       0.00      0.00      0.00         1\n\n    accuracy                           0.50         2\n   macro avg       0.25      0.50      0.33         2\nweighted avg       0.25      0.50      0.33         2\n'
```

In [65]:
```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(12,4))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=['No','Yes'],
    filled=True
)
plt.show()
```



In [ ]:

In [ ]: